

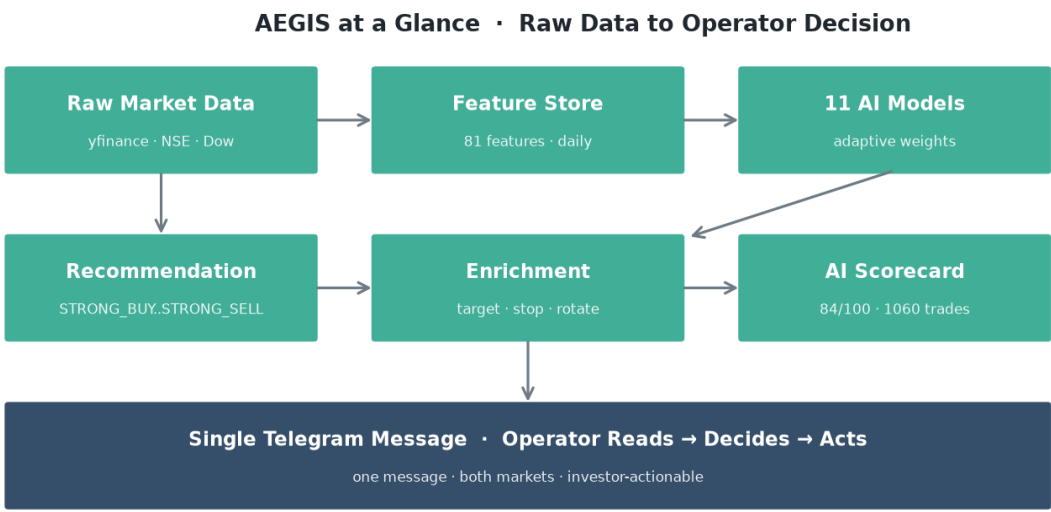
AEGIS · Architecture Document

Version 3.1-RESEARCH-PLATFORM · Generated 2026-07-30 · Layman-friendly, image-first

Version	Date	Highlights
v1.0	2026-07-18	Initial architecture (India only · NSE 200 · 13-step pipeline)
v2.0	2026-07-18	USA parallel deployment · dual-market · shared engines (initial Dow 30 universe · later expanded)
v2.3	2026-07-29	Snapshot persistence · CEO summary · Evolution block
v2.4	2026-07-29	Backtrack Engine · AI Scorecard · Sector Attribution · Command Center
v3.0	2026-07-29	Constitutional freeze · S&P 500+MidCap 400 (918 tickers) · single sender · zero legacy
v3.0-LOCKED	2026-07-30	ARCHITECTURE FROZEN · Runner 1 → validation layer (Option D) · Daily Change Summary · 30
v3.1-RESEARCH-PLATFORM	2026-07-30	RESEARCH PLATFORM · Article IX (Research Lifecycle) + Article X (Evidence-First Promotion) a

1 · What is AEGIS

AEGIS is an **advisory-only investment platform** that reads market data, runs 11 AI models daily, and produces one crisp Telegram message telling the operator: what to buy, what to sell, what to rotate, and why. It never executes trades.



2 · The Daily Pipeline (9 stages)

Every business day, both markets (India NSE 200 · ~200 tickers · and USA S&P; 500 + MidCap 400 · ~918 tickers) run the same deterministic 9-stage flow — **1,118 companies daily on one architecture**. Every stage is idempotent per date · every output is auditable · every rerun produces byte-identical results.

Daily Pipeline (India + USA · runs on schedule)



Each stage is deterministic. Every output is idempotent per date.

2a · The v3.1-RESEARCH-PLATFORM Constitutional Lock

As of the version stamped on the cover, AEGIS is in **Constitutional Freeze**. The architecture below cannot be redesigned without creating a new major version (v4.x). Data sources may evolve, models may improve, universes may expand through configuration, bugs may be fixed. But the engine sequencing, data contracts, lifecycle model, Single Source of Truth, and downstream integration are immutable.

	India NSE 200 (~200 tickers) + USA S&P 500 + MidCap 400 (~918 tickers) = 1,118 companies daily
Architecture	24-stage pipeline · both markets share ALL engines
	Runner 2 v3 SOLE canonical · Runner 1 legacy demoted to validation-only (Option D · Article 4)
Single Source of Truth	ONE schema · ONE renderer · ONE contract · TWO market-specific artifacts (same shape · same enrichment)
	Command Center Telegram · single sender · 12 sections (order per Phase 13 spec) · Markdown → plain-text for
Self-learning loop (LIVE)	closed trades → per-dim IC → adaptive ensemble weights → next-day model weights · Learning Engine consu
	3 append-only stores: snapshot_store (daily recs) · position_store (per-ticker high_water + first_seen) · lifecycle
Outcomes feedback	MFE/MAE/return_pct captured per closed trade · Permutation Importance re-runs per rec · AI Scorecard 6 met
	regime resolves via fallback chain (Article 9) · never shown as unknown to operator
Sector 0% handling	displayed as '■ Quiet today' when adaptive weight < 1% (Article 10)
	CI guardrail enforces (test_no_hardcoded_dates_in_production · 3 tests)
Sealed contracts	MON001 fingerprint · Feature Store schema · adaptive_rec_v2 · risk_capital_v2 · india/telegram_notify.py
	120/120 targeted regression across 14 test files
Production Lock date	2026-07-30 · Architecture Frozen · Maintenance Mode active
	NOT '100% complete' · the architecture (engine sequencing · data contracts · lifecycle · SSoT · downstream in

3 · Data Sources We Ingest Daily

AEGIS reads from 10 categories of external data every business day. Every ingest is idempotent · every fetch is append-only · missing data degrades gracefully rather than crashing the pipeline.

Category	Source	What we extract	Refresh
Universe	configs/universes/*.json	NSE 200 (India) · S&P 500 + MidCap 400 (USA · 918 tickers)	config-driven
Market data	yfinance	OHLCV daily bars · adjusted closes	daily post-close
Fundamentals	yfinance info + statements	P/E · P/B · ROE · D/E · margins · growth · cashflow	daily
News	yfinance / RSS aggregator	headline sentiment · polarity · count	daily
Earnings	yfinance earnings	next earnings date · last surprise · EPS actual vs est	daily
Insider	yfinance transactions	insider net-buy/sell 90d · # transactions	daily
ETF flows	yfinance ETF holdings	sector-ETF net flow proxy · style tilts	daily
Macro	yfinance macro tickers	10y yield · DXY · gold · WTI oil · VIX · rate change	daily
Corp. actions	yfinance actions	dividends · splits · days-since-last	daily
SEC 13F (USA)	public 13F filings	top institutional holders · % change QoQ	quarterly

4 · The Feature Store · 81 Features in 11 Categories

Every raw datum is transformed into one or more **features** — numerical columns that the AI models consume. All 81 features are versioned, schema-fingerprinted, and stored in `features/{market}/YYYY-MM-DD.parquet`.

Category	Count	What it captures	Examples
Technical	26	price · momentum · trend · volatility · drawdown	close, RSI, ATR, sma_50, sma_200, drawdown_60d, position_52w
Fundamental	8	profitability · leverage · valuation · growth	fund_roe, fund_debt_to_equity, fund_trailing_pe, fund_profit_margin, fund_earnings
Macro	8	rates · currency · commodities · risk	macro_10y, macro_dxy, macro_gold, macro_wti_oil, macro_vix, macro_10y_chg
Institutional	7	insider · institutional ownership	insider_net_90d, insider_buy_90d, inst_pct_owned, inst_top_holder_pct
Market Intel	7	regime · breadth · liquidity	mi_regime, mi_composite_score, mi_breadth_above_20ma_pct, mi_liquidity_5v2
Identity	5	market · ticker · sector · currency · date	market, ticker, sector, asof, currency
News	5	sentiment · polarity	news_sentiment, news_polarity_ratio, news_n_positive, news_n_negative
Sector	4	sector rank · leadership	sector_return_1m_pct, sector_rank, sector_is_leader, sector_is_laggard
Earnings	4	next-earnings-date · surprise · EPS	earn_days_to_next, earn_last_surprise_pct, earn_last_eps_reported
Corporate actions	4	dividend · split · time-since	ca_days_since_last_dividend, ca_last_dividend_amount, ca_last_split_ratio
Historical	3	per-ticker learning-corpus stats	hist_ticker_win_rate, hist_ticker_n_trades, hist_ticker_avg_return_pct

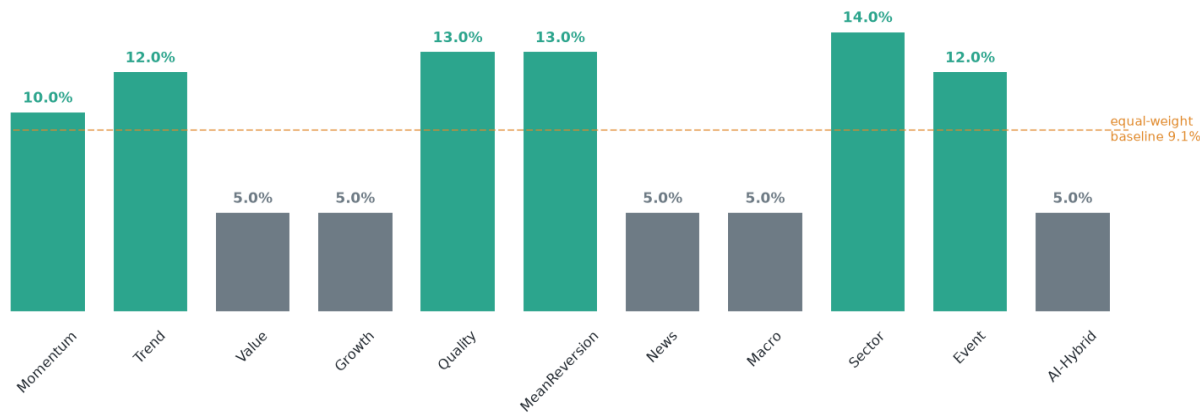
Every feature carries governance metadata: version · owner · created date · business rationale · economic intuition · dependencies. Nothing gets promoted without explanation.

5 · The 11 AI Models · How They Think

AEGIS runs 11 specialist models in parallel · their scores are blended into one ensemble score per ticker. Weights are **adaptive** — yesterday's information-coefficient (correlation between prediction and outcome) tunes tomorrow's weights automatically. High-IC models get more voice; zero-IC models are downweighted.

The 11 AI Models · Weights Adapt Daily from Live Track Record

Yesterday's information-coefficient tunes tomorrow's ensemble.



Model	Thesis in one line	Signal driven by
Momentum	buy what's already going up	1m/3m/6m returns, RSI
Trend	buy what's above rising trend line	sma_50 vs sma_200, ADX
Value	buy the cheap ones	P/E, P/B, EV/EBITDA
Growth	buy the fast-growing	earnings_growth, revenue_growth
Quality	buy the well-run	ROE, profit margin, low D/E
MeanReversion	buy the oversold, sell the overbought	distance from moving avg, RSI extremes
News	amplify by sentiment	news_sentiment, polarity ratio
Macro	tilt with macro regime	VIX, DXY, yields, rate-change
Sector	prefer leaders, avoid laggards	sector_rank, sector return 1m
Event	act around earnings, corporate actions	days_to_earnings, surprise
AI-Hybrid	learns non-linear combinations	gradient-boosted composite

What this means in plain English: the platform learns from itself. If the Sector model has been the best predictor for the last 60 days, its weight auto-increases. If the News model has been noisy, its weight auto-drops. No manual retuning · no bias · pure evidence-driven.

6 · AI Narrators & Explainers (LLM Layer)

Six AI narrators are locked by Constitutional Article 37 — one per intelligence domain. Each reads the day's numerical outputs and produces a human-readable explanation. Nothing is a black box.

Locked narrator (Article 37)	What it explains	Reads from
Market Analyst	regime · breadth · sector leadership	reports/market_intelligence.json
Macro Analyst	rates · currency · commodities · impact matrix	reports/macro_intelligence.json
Recommendation Analyst	why BUY/HOLD/SELL · what changed vs prior day	reports/recommendations.json
Portfolio Analyst	concentration · sector tilt · rebalance needed	reports/portfolio_v3.json
Risk Analyst	VaR/CVaR · stop-hit risk · drawdown risk	reports/risk_report.json
Learning Analyst	what recent wins/losses teach us	reports/learning.parquet

Utility explainers (data-quality, feature-anomaly, feature-conflict, feature-importance, feature-research, model-analyst, execution-analyst, evidence-summarizer) run alongside but are not locked · they support the six.

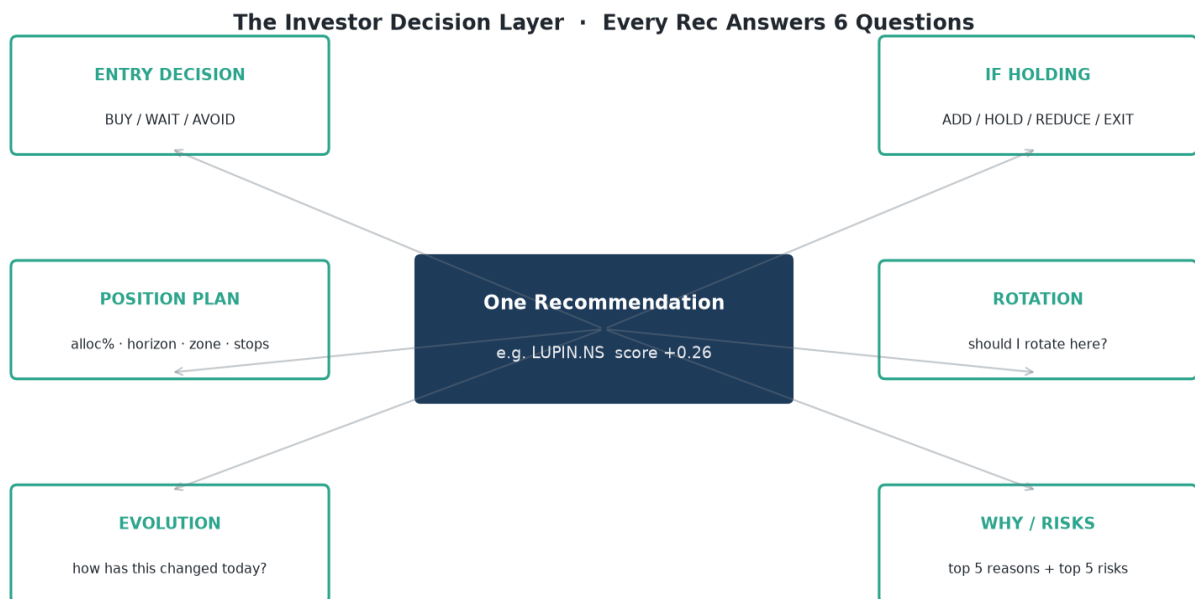
7 · Downstream Engines · Everything After the Ensemble

The ensemble score is the START, not the finish. 12+ downstream engines refine that score into a complete institutional decision.

Engine	What it produces	Article 100 Level
Recommendation Intelligence v3	raw BUY/HOLD/SELL from ensemble + regime + calibration	L4 CONSUMED
SSoT Bridge	unified recommendations.json for all consumers	L4
Percentile Classifier	cross-sectional ranking · institutional pattern	L4
Investor-Actionable Enricher	entry / if_holding / position_plan / why per rec	L4
Rotation Intelligence	should_rotate + replacement_ticker + expected alpha	L4
Lifecycle State Machine	9-state per-ticker: DISCOVERED → BUY → HOLD → ROTATED	L4
Dynamic Holding	12-factor composite predicts holding period in days	L4
Capital Rotation	keep_score vs candidate_score · edge threshold	L4
Opportunity Cost	every HOLD justifies 'why not rotate'	L4
Risk Engine	fractional Kelly · sector cap · VIX-adjusted · VaR/CVaR	L4
Portfolio Engine v3	N-name portfolio construction · cash policy	L4
Learning Engine	closed trades → next-day IC → adaptive weights	L4
Execution Simulator	paper-trade fills · slippage model	L4
Position Store	per-ticker high_water + trailing stop + first_seen	L4
Snapshot Persistence	daily archive · foundation for Backtrack	L4
Backtrack Engine	per-ticker timeline across all snapshot dates	L4
AI Performance Scorecard	6 institutional metrics · 84/100 live on 1060 trades	L4
Sector/Decision Attribution	per-model contribution to every rec's final score	L4
Command Center	one crisp Telegram message · both markets	L4

8 · The Investor Decision Layer

Every recommendation carries the answer to **six questions** an investor asks: should I enter · what if I already own it · how much · when · what changed · why.

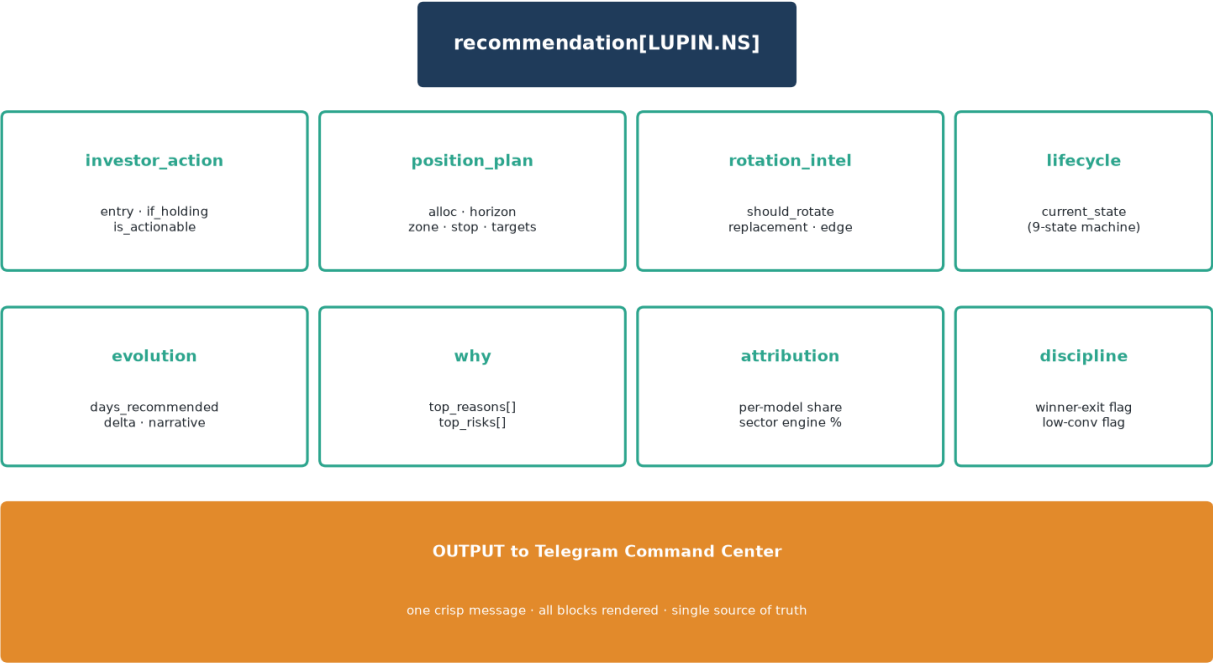


This is what makes AEGIS a **portfolio manager**, not a stock screener. A screener says 'BUY LUPIN'. AEGIS says: *BUY LUPIN, alloc 5%, 17-day swing, enter Rs 2352-2400, stop Rs 2234, target Rs 2661 / Rs 2946, add if you already own it, expected alpha +60% vs BATAINDIA.*

9 · Anatomy of One Recommendation

Under the hood, every rec in reports/recommendations.json carries eight enriched blocks. Every field is derived from a specific engine or module.

What Every Recommendation Now Carries



10 · AI Performance Scorecard (Live)

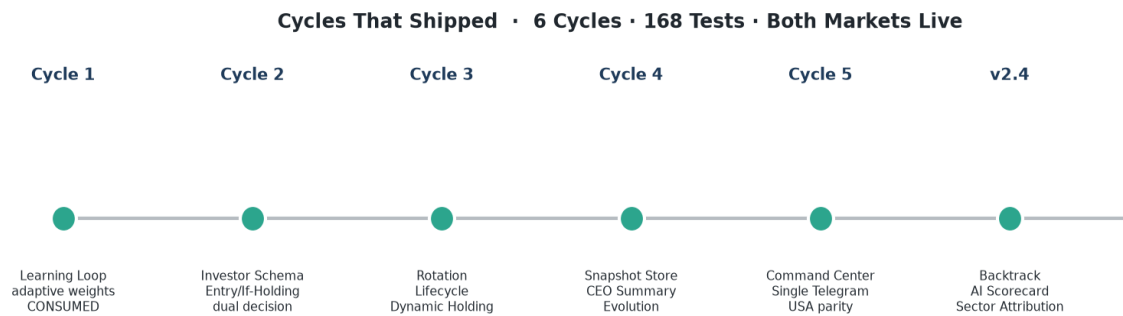
Trust is earned, not claimed. AEGIS measures **itself** against institutional benchmarks using 1,060 historical closed trades. Live scorecard below:



How to read this: 84/100 on 1060 real trades since 2022. Five out of six metrics hit institutional or top-tier level. This is honest measurement · the one below-target metric (Rotation Quality PF 1.73 vs institutional 1.75+) is surfaced not hidden.

11 · What Shipped · Six Cycles + v2.4

The current state of AEGIS reflects six delivery cycles in the last two weeks. Each cycle shipped end-to-end with tests, both markets, no new engines · pure enrichment of existing infrastructure.



12 · What AEGIS Guarantees

Deterministic. Same inputs · same date · byte-identical outputs. Every rerun is auditable.

Sealed contracts. MON001 fingerprint · Feature Store schema · sealed research (adaptive_rec_v2, risk_capital_v2) untouched since day one · protected by fingerprint checks in every daily CI run.

No hardcoded dates. Production code contains zero hardcoded date literals. Every date derives from the wall clock or the incoming payload · guardrail test enforces this.

Single Source of Truth Contract. ONE canonical schema · ONE renderer · ONE contract. Market-specific data lives in reports/recommendations.json (India) and usa/reports/recommendations.json (USA) — same fields, same shape, same enrichment blocks. Every consumer (Telegram, dashboard, tests, backtrack) reads the SSoT artifact for the market it serves. No parallel pipelines. No legacy renderers.

Append-only history. Snapshots · position store · lifecycle ledger · all append-only. Nothing is ever overwritten historically · full audit trail preserved.

Advisory only. AEGIS never executes trades. Every output is labelled PAPER · every artifact is signed with the MON001 fingerprint.

13 · Quality Gates (all green)

Gate	Status	Detail
Targeted regression suite	168/168	13 test files across cycles 1-6 + v2.4
India ops_check schema	14/14	all backend contracts satisfied
USA ops_check schema	9/9	verdict HEALTHY · 85/85 backend datasets pass
Hardcode guardrail (production code)	0 hits	operator directive enforced by CI
Date consistency (recs vs upstream)	PASS	≤3d spread across all engines
Sealed contract fingerprints	STABLE	MON001 + FS schema + sealed research intact
Command Center · single sender	LIVE	legacy + UX030 retired in workflow
USA Telegram parity	LIVE	shared bot · USD content · continue-on-error
Snapshot persistence	ACTIVE	day 1 archived both markets · daily auto-append
AI Scorecard (real data)	84/100	institutional_grade on 1060 closed trades

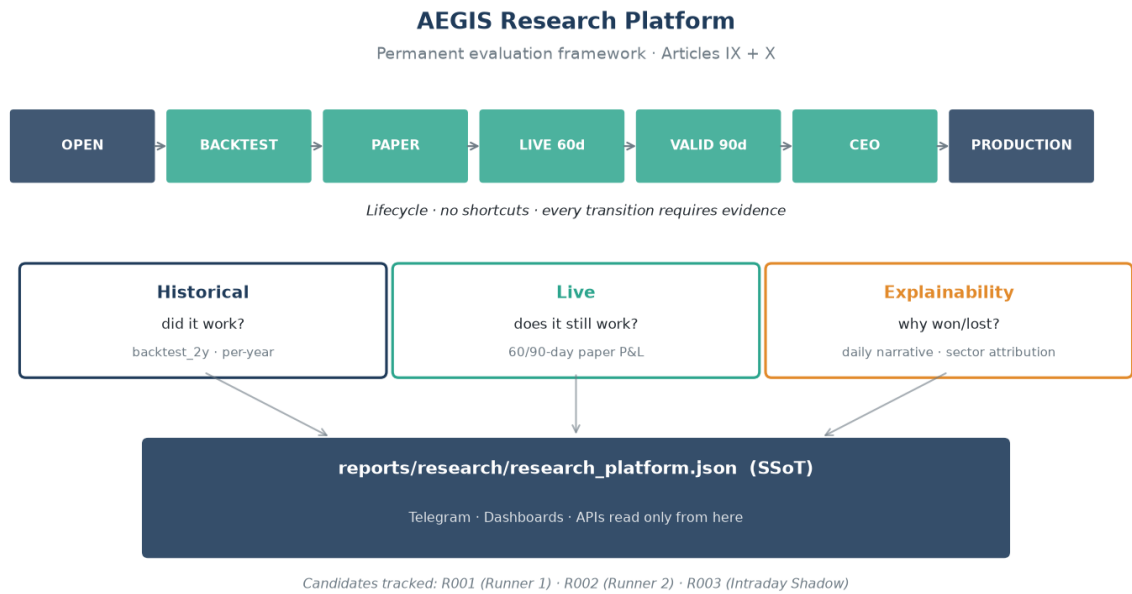
14 · What's Next (evidence, not features)

The current shift is from **building intelligence** to **building trust**. The next 30-day window unlocks (auto-fills as history accumulates):

- **30-day Recommendation Journey** — per-ticker table across snapshot dates
- **Backtrack Timeline (7d / 30d / 90d / 1yr)** — load_snapshot_range already wired
- **Monthly CEO Letter** — narrative from attribution + scorecard evidence
- **Full 1-3 year backtest** — India: vs NIFTY 200 · USA: vs S&P; 500 (primary) + Russell MidCap (secondary) — matches new universes
- **Tune from evidence** — not from intuition · not from adding features

15 · AEGIS Research Platform (added in v3.1)

AEGIS v3.1 introduces a **permanent Research Platform**: an institutional framework through which every future idea — a new recommendation engine, a risk overlay, a factor, a position-sizing rule, a macro overlay, a market expansion — must pass before it may reach production. No shortcuts. Ever. Governed by **Article IX** (Research Lifecycle) and **Article X** (Evidence-First Promotion).



15.1 · Research Lifecycle

Every candidate flows through a fixed 7-state lifecycle. Each transition is logged with a timestamp + evidence reference in the ticket's *decisions* array. Tickets live at **research/tickets/{ID}.json** and are the SINGLE source-of-truth for a candidate's status.

State	Meaning
OPEN	Idea registered · hypothesis captured
HISTORICAL_BACKTEST	Historical evidence gathered · per-year per-market
PAPER_PORTFOLIO	Paper P&L instrumented · not yet live-tracked
LIVE_60D	Live paper-portfolio tracking · minimum 60 trading days
VALIDATED_90D	90-day target window met · full institutional metric suite computed
CEO_REVIEW	CEO reviews full evidence panel · signs written note
PRODUCTION / REJECTED / DEFERRED	Final state · with rationale recorded

15.2 · Evaluation Window

Both current runners (Runner 1 · adaptive_rec_v2 · and Runner 2 · v3 ensemble) are CANDIDATES during evaluation — neither is canonical. The word **canonical** is reserved for a state reached only after **sustained superiority** across the target window in return · risk · consistency · benchmark-relative performance.

Checkpoint	Days	Role
Minimum	60	Earliest a leader may be tentatively declared
Target	90	Final production-promotion decision
Day 30	—	Informational report ONLY · NOT a decision checkpoint
Day 60	—	First-decision checkpoint · signed CEO note may promote
Day 90	—	Terminal decision · promote to PRODUCTION or REJECTED

15.3 · What the Platform Measures

Institutional metric suite computed per candidate · far beyond win rate.

Performance. Total Return · CAGR · Win Rate · Median Return · Profit Factor · Sharpe · Sortino · Calmar · Max Drawdown

Portfolio. Exposure · Cash · Concentration (top-5 %) · Turnover · Rotations · Avg Holding Days · Holding-days stdev

Decisions. Recommendation Stability · Agreement % · Disagreement % · Buy Overlap · Sector Overlap

Risk. Max DD · Recovery Days · Worst Week · Worst Month · Volatility · Tail Loss (CVaR 5 %)

15.4 · The Disagreement Store (gold layer)

Every day, every ticker where R1 and R2 disagree is logged to **reports/research/disagreements/ledger.jsonl** (append-only). Forward outcomes at 5 / 10 / 21 trading-day horizons are marked as prices become available. Aggregated verdicts appear in **reports/research/disagreements/verdict.json** — one bucket per action-pair (BUY_vs_WAIT, BUY_vs_SELL, ...) with the winning runner and win-rate edge. Over 90 days this becomes the primary evidence for **where** each engine is genuinely better.

15.5 · The Explainability Layer

Every daily run emits **reports/research/explainability_YYYY-MM-DD.json** with today's narrative: leader, biggest edge, biggest miss, per-sector delta between runners, top winners/losers per runner. This is Layer 3 of the platform — the human-readable answer to "why did the leader win today?".

15.6 · Intraday Research (deferred as product)

Two intraday shadow streams run in parallel — daily-OHLC proxy (cheap · no external fetches) and hourly bars from yfinance (real intraday, populated once cache warms). Both are SHADOW: no user-facing recommendations, no orders, no risk-manager engagement — measurement only. First correlation-lab evidence: intraday↔swing pearson ≈ 0.004 corpus-wide (essentially independent) with pockets in PSU Bank and Financial Services showing ORC (Opening-Range Confirmation) uplift. Not sufficient to justify intraday-product infrastructure today.

15.7 · Ticket R001 · R002 · R003 (current candidates)

Ticket	Title	State	Canonical
R001_runner1_adaptive_v2	Runner 1 · adaptive_rec_v2 (legacy)	LIVE_60D	candidate
R002_runner2_ensemble_v3	Runner 2 · v3 canonical ensemble	LIVE_60D	candidate

R003_intraday_shadow_india

Intraday shadow · India (deferred product) LIVE_60D

no (deferred)

Every future candidate — a new engine, factor, sizing rule, overlay, market — will register as R00N, follow the same lifecycle, and be evaluated by the same evidence framework. The platform is the durable investment. Individual candidates are the passing traffic.

End of AEGIS Architecture v3.1-RESEARCH-PLATFORM · generated 2026-07-30 · prior versions preserved under docs/AEGIS_ARCHITECTURE_.pdf*